

Worksheet 2

Student Name: Navjot Kaur

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1. Aim/Overview of the practical:

Interface a buzzer, led, pushbutton (**Pull Down**) and LCD with Arduino Uno. Use two led and two pushbuttons. Use pushbutton as toggle switch.

- Case 1: When press button 1 once then buzzer will turn **ON** and print “**Button1 ON**” in Row 2 and “**Buzzer ON**” in Row 1.
- Case 2: When we press button 1 again then buzzer will turn **OFF** and print “**Button1 OFF**” in Row 2 and “**Buzzer OFF**” in Row 1.
- Case 3: When we press button 2 once then LED will turn **ON** and print “**Button2 ON**” in Row 2 and “**Lights ON**” in Row 1.
- Case 4: When we release button 2 again then LED will turn OFF and print “**Button2 OFF**” in Row 2 and “**Light OFF**” in Row 2.

2. Apparatus (For applied/experimental sciences/materials based labs):

Hardware Requirements

- Arduino Uno R3
- Breadboard
- Push Button
- LED
- Resistor (220Ω)
- LCD Display (16*2 I2C)
- Jumper Wires
- USB Cable/Power Supply
- Buzzer

Software requirements

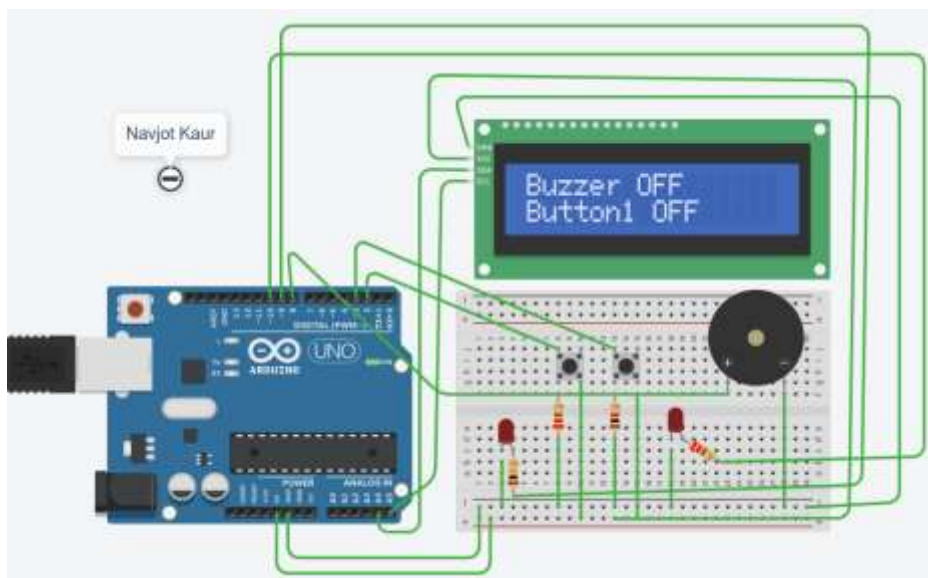
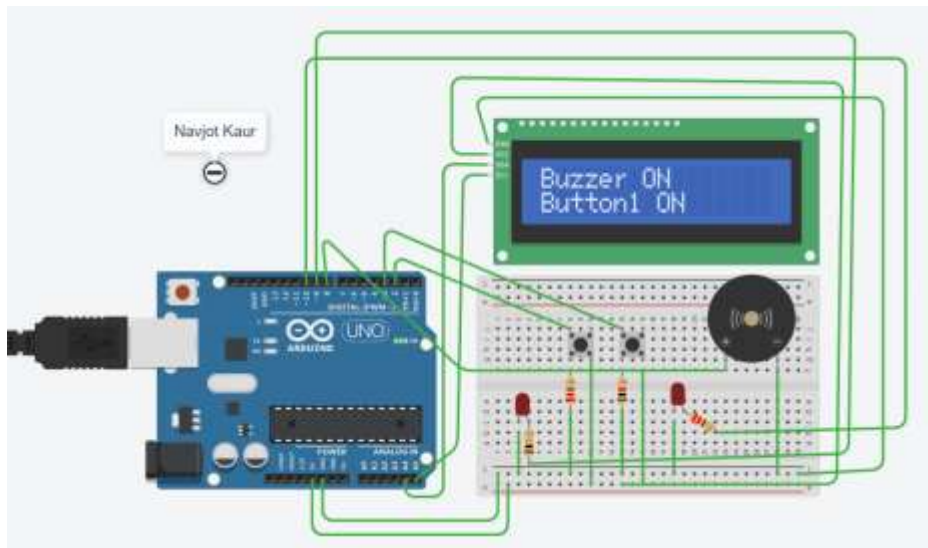
- Arduino IDE
- Tinkercad Circuits

3. Circuit Diagram (TinkerCad):

a) Case 1: When press button 1 once then buzzer will turn ON and print “Button1 ON” in Row 2 and “Buzzer ON” in Row 1.

b) Case 2: When we press button 1 again then buzzer will turn OFF and print “Button1 OFF” in Row 2 and “Buzzer OFF” in Row 1.

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);
int b1 = 2;
int buzzer = 8;
int buzzerState = 0;
void setup() {
  pinMode(b1, INPUT);
  pinMode(buzzer, OUTPUT);
  lcd.init();
  lcd.backlight();
  lcd.print("System Ready");
  delay(1000);
  lcd.clear();
}
void loop() {
  if (digitalRead(b1) == HIGH) {
    buzzerState = !buzzerState;
    if (buzzerState == 1) {
      tone(buzzer, 2000);
      lcd.clear();
      lcd.setCursor(0, 0);
      lcd.print("Buzzer ON");
      lcd.setCursor(0, 1);
      lcd.print("Button1 ON");
    }
    else {
      noTone(buzzer);
      lcd.clear();
      lcd.setCursor(0, 0);
      lcd.print("Buzzer OFF");
      lcd.setCursor(0, 1);
      lcd.print("Button1 OFF");
    }
    delay(300);
  }
}
```



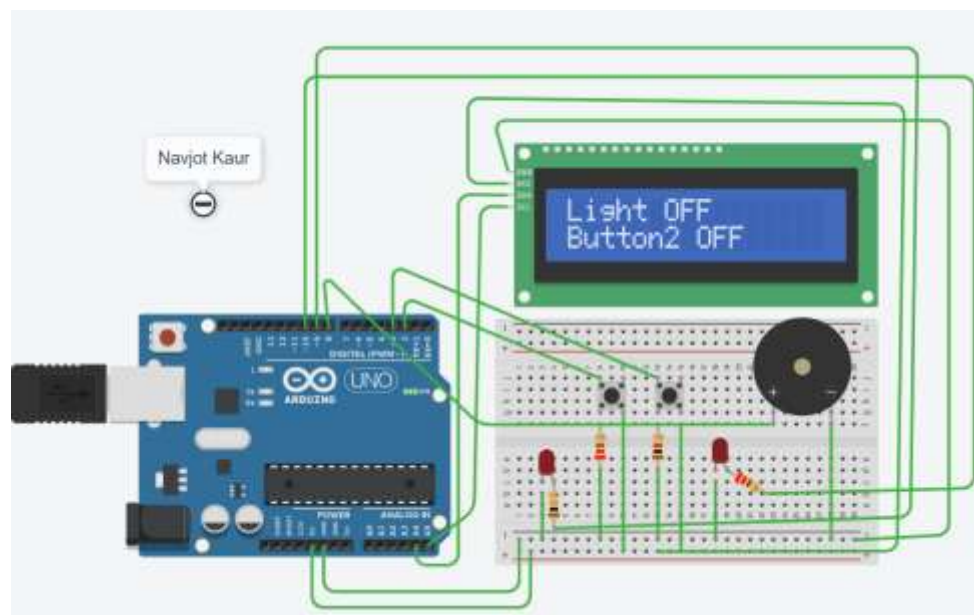
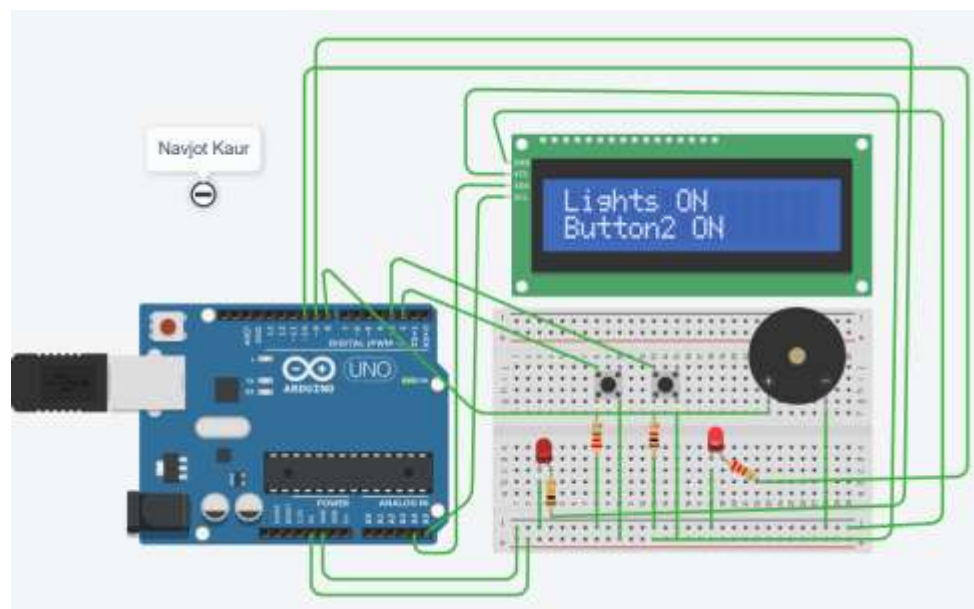
c) Case 3: When we press button 2 once then LED will turn ON and print “Button2 ON” in Row 2 and “Lights ON” in Row 1.

d) Case 4: When we release button 2 again then LED will turn OFF and print “Button2 OFF” in Row 2 and “Light OFF” in Row 2.

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);
int b2 = 3;
int led1 = 9;
int led2 = 10;
int lightState = 0;
```



```
void setup() {  
  pinMode(b2, INPUT);  
  pinMode(led1, OUTPUT);  
  pinMode(led2, OUTPUT);  
  lcd.init();  
  lcd.backlight();  
  lcd.print("System Ready");  
  delay(1000);  
  lcd.clear();  
}  
void loop() {  
  if (digitalRead(b2) == HIGH) {  
    lightState = !lightState;  
    if (lightState == 1) {  
      digitalWrite(led1, HIGH);  
      digitalWrite(led2, HIGH);  
      lcd.clear();  
      lcd.setCursor(0, 0);  
      lcd.print("Lights ON");  
      lcd.setCursor(0, 1);  
      lcd.print("Button2 ON");  
    }  
    else {  
      digitalWrite(led1, LOW);  
      digitalWrite(led2, LOW);  
      lcd.clear();  
      lcd.setCursor(0, 0);  
      lcd.print("Light OFF");  
      lcd.setCursor(0, 1);  
      lcd.print("Button2 OFF");  
    }  
    delay(300);  
  }  
}
```



5. Learning outcomes (What I have learnt):

1. Learned to interface pushbuttons, LEDs, buzzer, and LCD with Arduino Uno.
2. Learned to use pushbuttons as toggle switches.
3. Learned to control LEDs and buzzer using Arduino.
4. Learned to display ON/OFF status on the LCD.
5. Learned basic digital input and output operations.